MTP PROJECT PRESENTATION

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TEAM MEMBERS

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INTRODUCTION

- **The Core Challenge:**
  - Financial institutions possess vast amounts of sensitive customer data.
  - While this data is invaluable for building **predictive models** (e.g., for credit risk, identifying fake transactions)
  - **Privacy regulations** and **confidentiality** make it impossible to share or use freely.
- **Our Objective:** This project aims to create a privacy-preserving generative model for financial data.
- **The Desired Outcome:**
  - Produce **high-quality synthetic dataset** that mirrors the statistical properties of the real data
  - But contains **no actual individual information**, enabling safe and effective model development.

PROBLEM STATEMENT

Privacy Preserving Generative Models:
This project explores the development of privacy-preserving generative models for financial data, aiming to enable the creation of synthetic datasets that preserve the statistical properties of real financial data while protecting sensitive individual-level information

LITERATURE REVIEW

- **Research Paper 1: PPGAN: PRIVACY-PRESERVING GENERATIVE ADVERSARIAL NETWORK [LINK]:**
  - PPGAN integrates Differential Privacy into Wasserstein GANs to prevent data leakage by adding Gaussian noise and clipping gradients.
  - It tracks privacy loss using a Moments Accountant, balancing privacy and data utility.
  - Experiments on MNIST and medical data show PPGAN produces high-quality synthetic data with strong privacy protection, outperforming other private GANs like DPGAN.

- **Research Paper 2: PATE-GAN: GENERATING SYNTHETIC DATA WITH DIFFERENTIAL PRIVACY GUARANTEES [LINK]:**
  - PATE-GAN replaces the standard GAN discriminator with an ensemble of teacher discriminators and a differentially private student discriminator, ensuring that no individual data sample significantly influences the model. The approach tightly bounds privacy loss while maintaining high data utility.
  - Experiments on multiple real-world datasets show that PATE-GAN consistently outperforms the prior state-of-the-art (DPGAN) in generating realistic, privacy-preserving data, achieving better predictive model performance and preserving ranking consistency across algorithms

- **Research Paper 3: CTAB-GAN+: Enhancing Tabular Data Synthesis [LINK]**
  - CTAB-GAN+ is an improved conditional GAN designed for generating realistic and privacy-preserving tabular data.
  - Across seven datasets, CTAB-GAN+ achieves up to 56% better data utility and 48% improvement under differential privacy compared to state-of-the-art methods.

DATASET DETAILS

1. **Dataset Source:** UCI Machine Learning Repository (Taiwan).
2. **Scope:**
   - Realistic proxy for financial data
   - Records for **30,000 customers** from April to September 2005.
3. **Features:** A rich mix of 23 predictive attributes, including:
   - Demographics: Age, Gender, Education.
   - Financial Standing: Credit Limit.
   - Historical Behavior: Six months of detailed payment statuses, bill amounts, and payment amounts.
4. **Prediction Goal:** A clear, binary target variable: **Will the client default on next month's payment?**

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19
ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_
1	20000	2	2	1	24	2	2	-1	-1	-2	-2	3913	3102	689	0	0	0	0	
2	120000	2	2	2	26	-1	2	0	0	0	2	2682	1725	2682	3272	3455	3261	0	
3	90000	2	2	2	34	0	0	0	0	0	0	29239	14027	13559	14331	14948	15549	1518	
4	50000	2	2	1	37	0	0	0	0	0	0	46990	48233	49291	28314	28959	29547	2000	
5	50000	1	2	1	57	-1	0	-1	0	0	0	8617	5670	35835	20940	19146	19131	2000	
6	50000	1	1	2	37	0	0	0	0	0	0	64400	57069	57608	19394	19619	20024	2500	
7	500000	1	1	2	29	0	0	0	0	0	0	367965	412023	445007	542653	483003	473944	55000	
8	100000	2	2	2	23	0	-1	-1	0	0	-1	11876	380	601	221	-159	567	380	
9	140000	2	3	1	28	0	0	2	0	0	0	11285	14096	12108	12211	11793	3719	3329	
10	20000	1	3	2	35	-2	-2	-2	-2	-1	-1	0	0	0	0	13007	13912	0	
11	200000	2	3	2	34	0	0	2	0	0	-1	11073	9787	5535	2513	1828	3731	2306	
12	260000	2	1	2	51	-1	-1	-1	-1	-1	2	12261	21670	9966	8517	22287	13668	21818	
13	630000	2	2	2	41	-1	0	-1	-1	-1	-1	12137	6500	6500	6500	6500	2870	1000	
14	70000	1	2	2	30	1	2	2	0	0	2	65802	67369	65701	66782	36137	36894	3200	
15	250000	1	1	2	29	0	0	0	0	0	0	70887	67060	63561	59696	56875	55512	3000	
16	50000	2	3	3	23	1	2	0	0	0	0	50614	29173	28116	28771	29531	30211	0	
17	20000	1	1	2	24	0	0	2	2	2	2	15376	18010	17428	18338	17905	19104	3200	
18	320000	1	1	1	49	0	0	0	-1	-1	-1	253286	246536	194663	70074	5856	195599	10358	
19	360000	2	1	1	49	1	-2	-2	-2	-2	-2	0	0	0	0	0	0	0	
20	180000	2	1	2	29	1	-2	-2	-2	-2	-2	0	0	0	0	0	0	0	
21	130000	2	3	2	39	0	0	0	0	0	-1	38358	27688	24489	20616	11802	930	3000	
22	120000	2	2	1	39	-1	-1	-1	-1	-1	-1	316	316	316	0	632	316	316	
23	70000	2	2	2	26	2	0	0	2	2	2	41087	42445	45020	44006	46905	46012	2007	
24	450000	2	1	1	40	-2	-2	-2	-2	-2	-2	5512	19420	1473	560	0	0	19428	
25	90000	1	1	2	23	0	0	0	-1	0	0	4744	7070	0	5398	6360	8292	5757	
26	50000	1	3	2	23	0	0	0	0	0	0	47620	41810	36023	28967	29829	30046	1973	
27	60000	1	1	2	27	1	-2	-1	-1	-1	-1	-109	-425	259	-57	127	-189	0	
28	50000	2	3	2	30	0	0	0	0	0	0	22541	16138	17163	17878	18931	19617	1300	
29	50000	2	3	1	47	-1	-1	-1	-1	-1	-1	650	3415	3416	2040	30430	257	3415	

Attributes of dataset

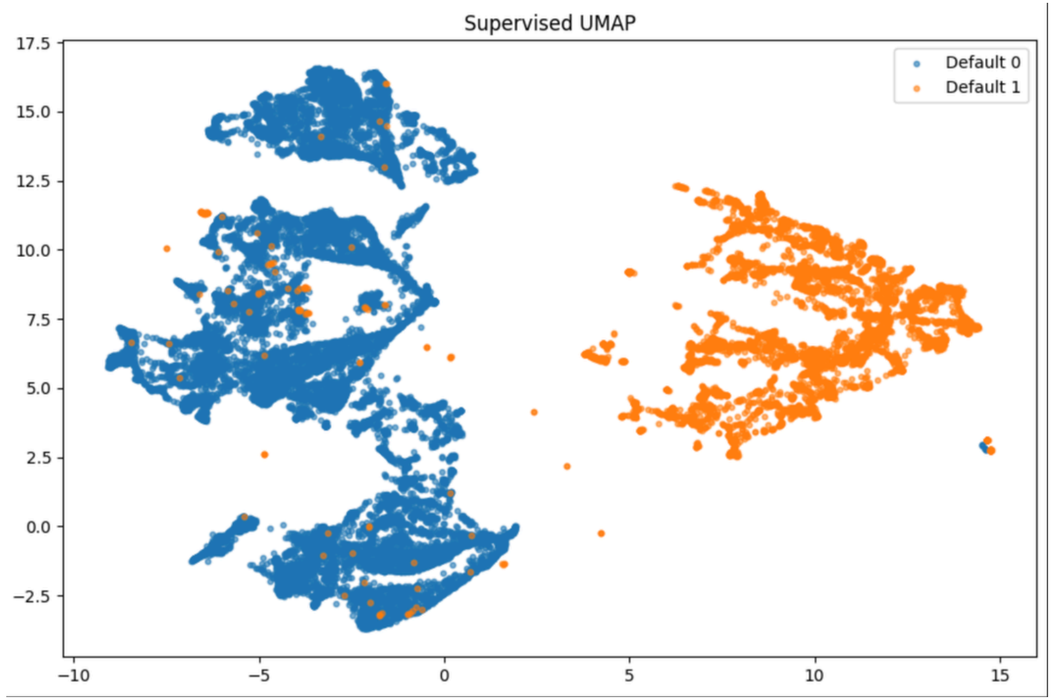
- **Timeline:** April 2005 - September 2005
- No of **users**: 30k
- **Target** Variable
  - 1 = Yes (client defaulted)
  - 0 = No (client did not default)
- **Payment History** (April–September 2005)
  - X6 - X11: Repayment status history
  - X6: September 2005
  - X7: August 2005
  - X8: July 2005
  - X9: June 2005
  - X10: May 2005
  - X11: April 2005
- Scale:
  - 1 = paid duly
  - 1 = delay 1 month
  - 2 = delay 2 months
  - 8 = delay 8 months
  - 9 = delay ≥ 9 months

Demographic Variables:

- X1: Amount of given credit (NT dollar)
  - includes individual and family/supplementary credit
- X2: **Gender**
  - 1 = male
  - 2 = female
- X3: **Education**
  - 1 = graduate school
  - 2 = university
  - 3 = high school
  - 4 = others
- X4: **Marital Status**
  - 1 = married
  - 2 = single
  - 3 = others
- X5: **Age**
  - years

ANALYSIS OF DATASET

- **Class Distribution:** imbalanced (~**22%** default rate)
- **Visual Analysis (t-SNE & UMAP):**
  - Partial separation but significant overlap between defaulters and non-defaulters
  - Data is complex and non-linear.



- **Predictive Quality (Modeling Benchmark):** We trained four classifier models to set a performance benchmark:
  - Logistic Regression
  - Random Forest
  - LightGBM
  - ANN
- **Key Finding:**
  - **ANN model** performed best in terms of AUC-ROC and accuracy.
  - Primary goal is to improve recall for the default class
  - Failing to predict a default (FN) is the most costly error.

Models	ROC-AUC	Accuracy
Random Forest	0.75	81.2 %
LightGBM	0.77	64.2 %
ANN	0.77	74.5 %

ERRORS

1. **False Positives** (Type I Error): Flagging a normal transaction as fraud.
2. **False Negatives** (Type II Error): Missing an actual fraud.

	Predicted Negative	Predicted Positive
Actual Negative	TN	FP
Actual Positive	FN	TP

- **Accuracy** = (TP + TN) / Total
- **Precision** = TP / (TP + FP) → "When I predict positive, how often am I right?"
- **Recall (Sensitivity, TPR)** = TP / (TP + FN) → "Of all actual positives, how many did I catch?"
- **Specificity (TNR)** = TN / (TN + FP) → "Of all actual negatives, how many did I catch?"
- **F1-score** = Harmonic mean of Precision & Recall.

- We have to reduce false negatives as we don't want to miss a fraud transaction.
- Our aim should be to increase recall, even if precision slightly drops

OUR APPROACH

Why CTAB-GAN+?

State-of-the-art generative model for **high-quality**, **privacy-preserving** tabular data synthesis, making it ideal for sensitive financial datasets.

Strengths:

- **Mixed Data Handling:** Supports continuous, categorical & mixed-type variables
- **Conditional Generation:** Effectively balances class distribution
- **Downstream-Aware:** Generates data optimized for ML model performance
- **Stable Training:** Leverages Wasserstein loss + Gradient Penalty for smooth convergence
- **Privacy by Design:** Integrates Differential Privacy (DP-SGD) to safeguard sensitive data
- **Optimal Trade-off:** Achieves high data fidelity with strong privacy guarantees

UNDERSTANDING THE LOSS FUNCTIONS

1. Adversarial Loss (WGAN-GP)

- Drives realism: Generator learns to fool the Discriminator.
- Uses **Wasserstein loss** + Gradient Penalty for stable convergence.

2. Information Loss

- Matches key statistics (mean, std, distribution shape) between real & synthetic data.
- Ensures statistical fidelity and preserves data structure.

3. Generator / Condition Loss

- Aligns generated data with requested conditions.
- Handles class imbalance and enhances conditional accuracy.

4. Downstream Loss

- Trains an auxiliary classifier/regressor alongside the GAN.
- Guarantees data is predictively meaningful for tasks like loan default or fraud detection.

Combined Objective function: L*(G)* = L(*adv)* + L(*info)* + L(*downstream)* + L(*cond)*

Each term contributes to:

- Realism → Adversarial Loss
- Statistical accuracy → Information Loss
- Task relevance → Downstream Loss
- Conditional control → Generator/Condition Loss

HOW WE PRESERVE PRIVACY

Core Idea:

We integrate **Differential Privacy (DP),** a mathematical guarantee ensuring that removing any single person's data does not significantly change the model's output.

How It Works (DP-SGD)

Privacy Accounting:

- Tracks total privacy loss using **Renyi Differential Privacy (RDP)**.
- Balances between privacy strength (ε) and data utility.
- $P[M(D) \in S] \leq (e^{\varepsilon}) * P[M(D') \in S] + \delta$
- D and D` are adjacent datasets

Noise Addition:

- Adds controlled Gaussian noise to clipped gradients.
- Masks individual contributions, creating statistical uncertainty.

Gradient Clipping:

- Limits how much influence each data point has on model updates.
- Prevents any one sample from dominating learning.

Result:

- Protects individuals from membership inference or **data reconstruction attacks**.
- Guarantees **mathematical anonymity:** No adversary can tell if someone's data was used.
- Enables **safe data sharing** in sensitive domains like finance & healthcare.

EVALUATION FRAMEWORK

1) Statistical Fidelity

- Kolmogorov–Smirnov **(K-S) Test →** Compares real vs. synthetic variable distributions.
- Jensen–Shannon Divergence **(JSD)** → Measures similarity for categorical variables.
- Wasserstein Distance **(WD**) → Evaluates continuous variable similarity.
- Pairwise Correlation Difference **(Diff. Corr.)** → Checks if feature relationships are preserved.
- Correlation **Heatmap** Comparison → Visual validation of statistical integrity.

2) Machine Learning Utility

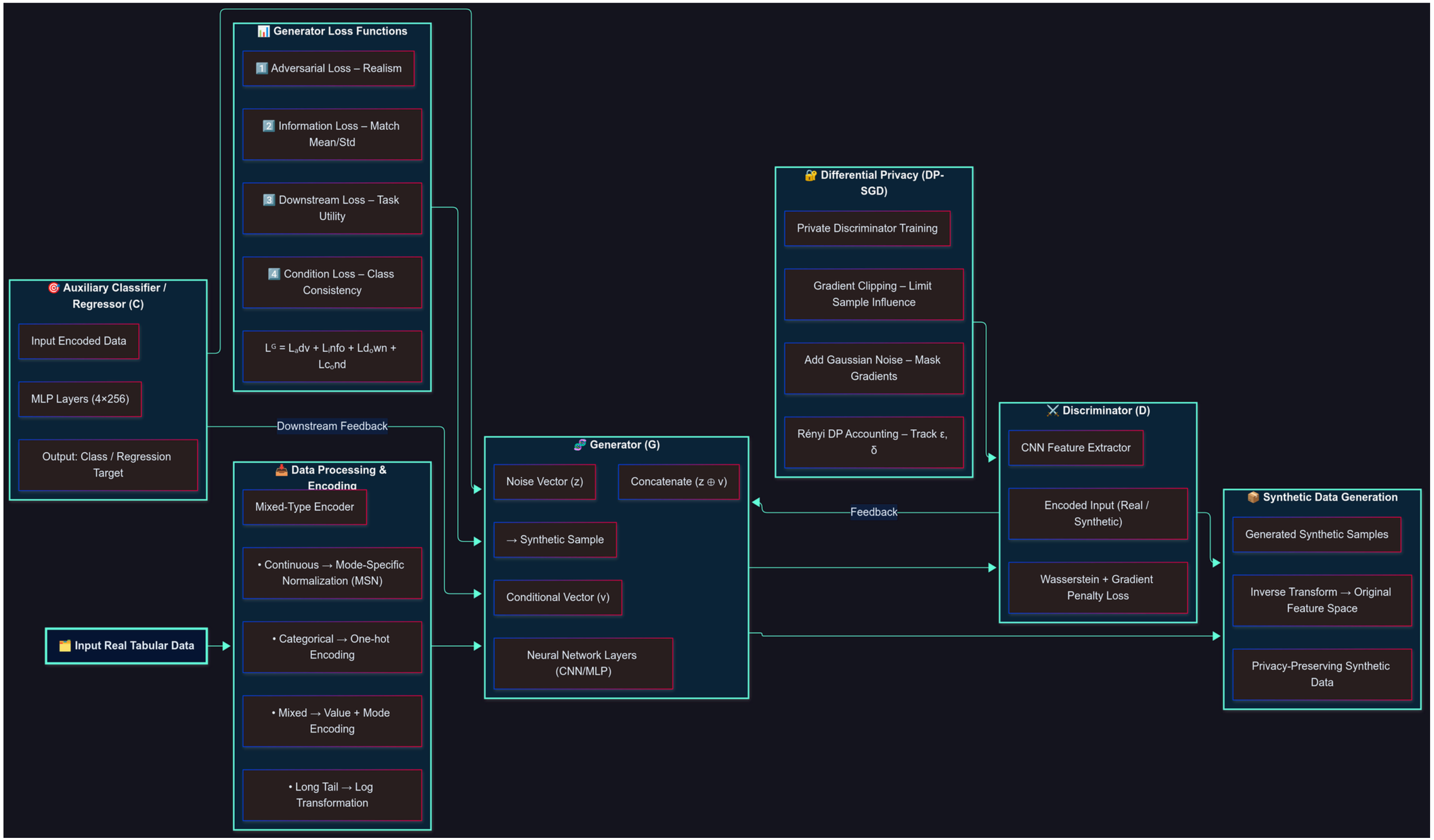
Measures how well synthetic data retains downstream ML performance.

- Train-Synthetic, Test-Real **(TSTR)** → Model trained on synthetic data, tested on real data.
- **Accuracy, F1-Score, AUC** → Classification performance metrics.
- **MAPE, EVS, R^2** → Regression task evaluation.

3) Privacy Assurance

- Distance to Closest Record **(DCR)** → ensures no synthetic record closely matches a real one.
- **ε-δ** Differential Privacy Budget → quantifies privacy guarantees.
- **Membership Inference Resistance** → tests protection from reverse engineering.

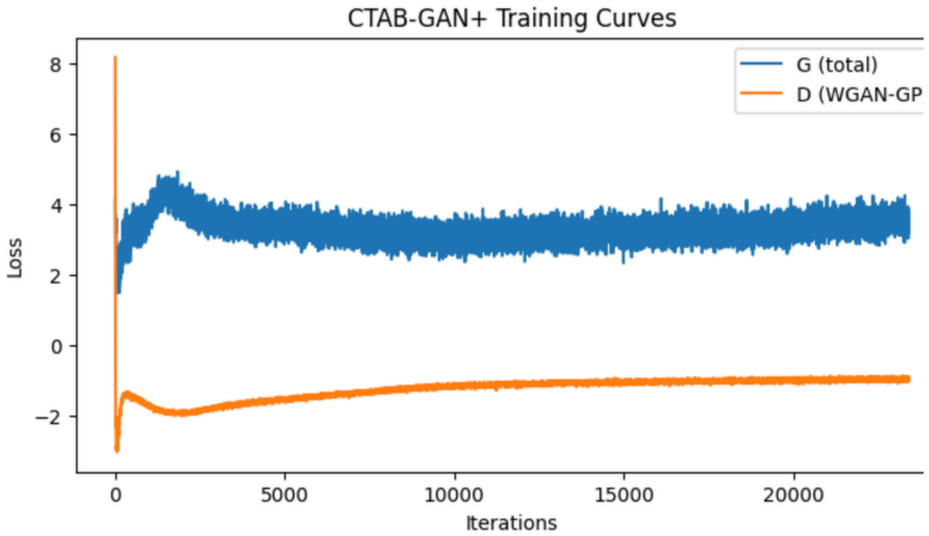
MODEL ARCHITECTURE



IMPLEMENTATION RESULTS

Without Differential Privacy

Training curve:



- The generator and discriminator losses stabilize quickly, showing smooth and **balanced training**.
- WGAN-GP ensures **stable convergence**, leading to realistic and high-quality synthetic data.

Evaluation Results:

```
========================================
1) Column-wise Distributional Similarity
========================================
Continuous (KS) summary: {'mean_ks': 0.17518333333333336, 'median_ks': 0.13296666666666668, 'n_cols': 24}
Categorical (JSD) summary: {'mean_jsd': None, 'median_jsd': None, 'n_cols': 0}

=============================================
2) Correlation Structure Similarity (Spearman)
=============================================
{'spearman_mae_upper': 0.07185055920621977, 'spearman_max_abs_err': 0.29356822717250103, 'n_pairs': 276}

========================================
3) Privacy (Distance to Closest Record)
========================================
{'mean_min_dist_synth_to_real': 1.3263235017288997, 'median_min_dist_synth_to_real': 1.051324048479524, 'pct_below_0p5_synth_to_real': 15.

=============================================
4) Detection (Real vs Synthetic Classifier AUC)
=============================================
{'detection_auc': 0.5802612222222223, 'ideal_random': 0.5}
```

- Column-wise Distributional Similarity **(K-S Test)**
  - Synthetic data closely follows real feature distributions.
- Correlation Structure **(Spearman)**
  - Strong preservation of relationships between variables.
- Privacy (Distance to Closest Record – **DCR**)
  - No synthetic record is identical or too close to real data — low re-identification risk.
- Detection (Real vs Synthetic Classifier **AUC**)
  - Model-generated data is indistinguishable from real samples.

Synthetic Sample:

Synthetic sample:

	ID	LIMIT_BAL	SEX	EDUCATION	MARRIAGE	AGE	PAY_0	PAY_2	PAY_3	PAY_4	...	BILL_AMT4	BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2
0	3878.620333	244085.032233	0.999862	0.999842	0.999858	50.398884	-1.001253	-1.000948	-1.000132	-1.000104	...	9751.654086	1866.722284	7547.119090	1.298754e+04	2447.108459
1	22171.604627	190574.838340	2.000032	1.999938	2.000063	28.507713	-0.000105	0.000474	-0.000309	0.000328	...	-79.731170	1224.844756	658.057588	5.319388e+03	3808.235734
2	15670.591615	81906.439975	1.000142	2.000312	1.000077	37.488571	0.000121	-0.000131	0.000101	0.000220	...	71564.301905	48718.675305	25229.565629	2.543793e+03	5073.007671
3	25766.444215	86030.089645	2.000185	3.000145	1.000311	50.250778	-0.000360	-0.000435	-0.000509	0.000607	...	28660.909029	29965.652970	30328.896283	1.985584e+03	1819.048681
4	17102.841502	151867.425445	1.999230	0.999110	0.999729	28.926159	0.998864	-2.003061	-2.003333	-2.002704	...	420.266844	1115.677661	-484.582189	-1.254732e-07	451.396766

5 rows × 25 columns

Machine Learning Utility Evaluation:

```
=== Random Forest Utility (Classification) ===
Train REAL → Test REAL (baseline): {'accuracy': 0.8148888888888889, 'f1': 0.4656831302116742, 'roc_auc': 0.7653399493038977, 'pr_auc': 0.5401074061619413}
Train SYNTH → Test REAL: {'accuracy': 0.821047619047619, 'f1': 0.4636026263202969, 'roc_auc': 0.7390254311052638, 'pr_auc': 0.5223961932388622}
```

Random Forest Utility (Classification)

- Model trained on synthetic data achieves performance close to the real-data baseline —
- Accuracy: 0.82 vs 0.81
- ROC-AUC: 0.74 vs 0.77.
- Indicates that CTAB-GAN+ preserves ML utility and generates high-quality, task-relevant data.

Evaluation Summary:

CTAB-GAN+ generates statistically **faithful, realistic**, and **privacy-safe synthetic tabular data**, even without applying explicit differential privacy.

PROJECT TIMELINE [DONE]

Week 1 (11/08/25 – 17/08/25):
Conducted a literature survey and explored various research papers. Experimented with alternative models to traditional GANs, such as PPGAN, PATE-GAN, and DP-GAN.

Week 2 (18/08/25 – 25/08/25):
Explored different financial datasets including the IBM Dataset, Kaggle Datasets, and the UCI Machine Learning Repository (Taiwan Credit Dataset).

Weeks 3 & 4 (26/08/25 – 10/09/25):
Performed dataset analysis and trained several classifier models such as Logistic Regression, Random Forest, and Artificial Neural Networks (ANN).

Week 5 (11/09/25 – 18/09/25):
Implemented the CTAB-GAN+ research paper, without differential privacy.

PROJECT TIMELINE [FUTURE]

Week 6 (06/10/25 – 13/10/25):

- Integrate Differential Privacy (DP) into the existing CTAB-GAN+ implementation using the Opacus library.
- Tune hyperparameters such as noise multiplier, clipping norm, and batch size to balance privacy and data utility.

Week 7 (14/10/25 – 21/10/25):

- Evaluate the impact of DP integration by comparing results with the non-DP model.
- Metrics: utility (accuracy, F1-score) vs privacy (ε-value, δ-value).

Weeks 8 & 9 (22/10/25 – 05/11/25):

- Start working with the Amex dataset in collaboration with the Amex team.
- Understand the data schema, preprocess features (handling categorical/numerical variables, normalization)
- Train CTAB-GAN+ with DP on the Amex dataset.
- Generate synthetic data and evaluate its statistical fidelity and privacy leakage using metrics like (Wasserstein Distance, Correlation Preservation, Membership Inference Risk) and comparing synthetic vs real data distributions.

Week 10 (06/11/25 – 13/11/25):

- Optimize model architecture and DP parameters for better synthetic data quality.
- Prepare evaluation results (tables, plots, and comparative summaries).
- Begin drafting a report summarizing key findings and results.

THANK YOU